

DRAWING TITLE

PIE

PRODUCT INTELLIGENCE ENGINE

ERP records the transaction. PIE understands the product and decides what should happen.

DETERMINISTIC PRODUCT INTELLIGENCE FOR B2B INDUSTRIAL DISTRIBUTION

4U PRECISION	HYDERABAD, INDIA	[STAGE]
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A quote begins as a sentence, not a part number

The request arrives unstructured. Everything after it is manual.



FRICITION POINTS

- | | | | |
|----|---|----|---|
| 01 | RFQ arrives — WhatsApp, email, PDF, photo | 04 | No grade field — expert memory fills it |
| 02 | Code decoded by hand | 05 | Cost and stock in another screen |
| 03 | Item master searched — 15,028 items, name is not identity | 06 | Price set from memory or last invoice |

ELAPSED: [TO VALIDATE]

ERP records transactions. It does not understand products.

WHAT ERP KNOWS

- SKU string
- stock on hand
- last cost
- last selling price
- customer
- tax code

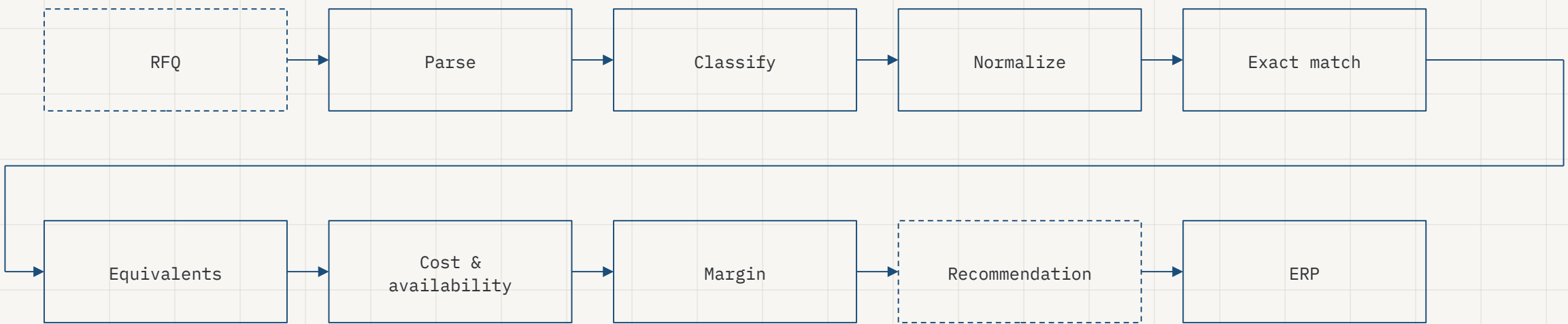
WHAT A QUOTE NEEDS

- what the product physically is
- whether another product substitutes
- whether this price holds margin
- whether this customer justifies it

Measured on this item master: 1.4% of item names carry a recognisable grade token.
Grade is the attribute the decision runs on.

The bottleneck is not recording the order. It is turning an unstructured request into a technically correct, commercially actionable quote — and that step lives in one person's head.

One pipeline, ten stages, one legend

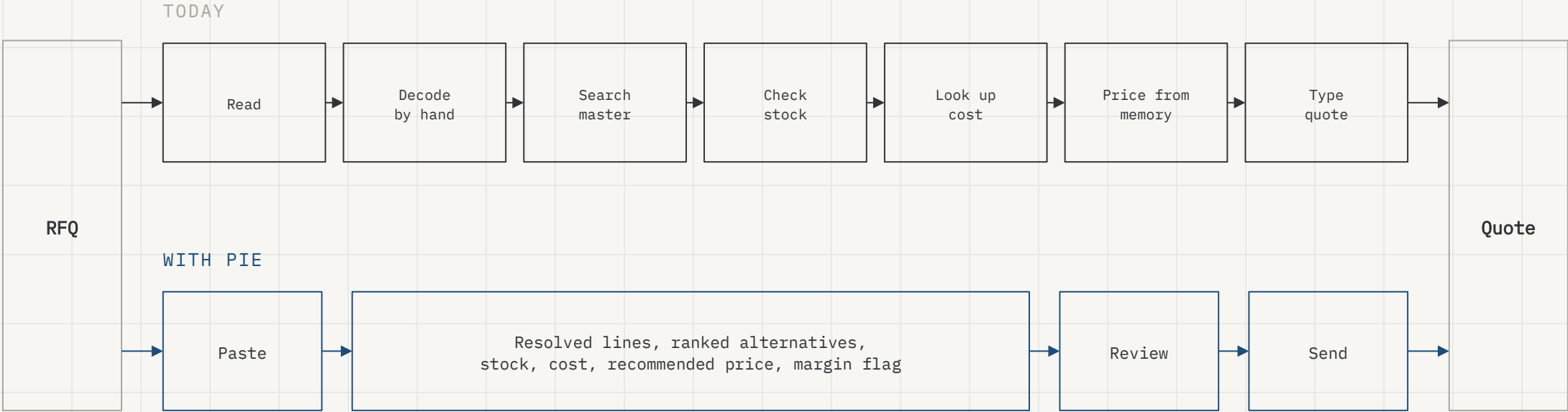


DETERMINISTIC

AI

Eight of the ten stages are deterministic. Only intake and explanation are dashed.
Every emitted field carries provenance, confidence, and a character span.

Same request in. Same quote out. Six steps removed.



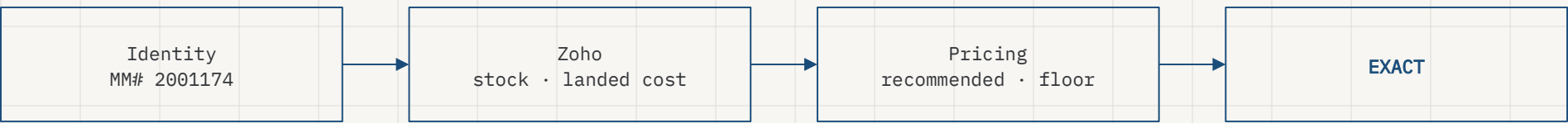
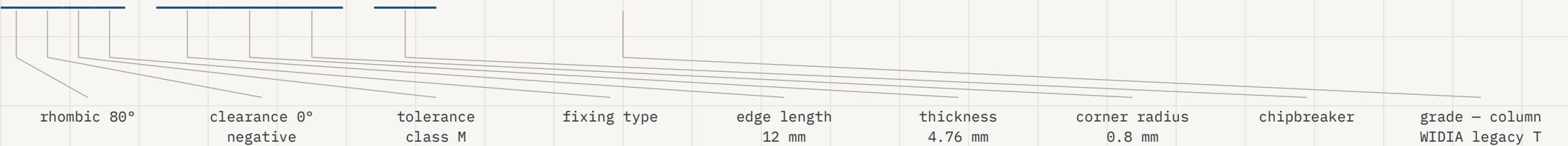
The human still approves every line. PIE removes the lookup, not the judgement.

[TO VALIDATE] MINUTES SAVED PER LINE

One line, end to end

LIVE ENGINE OUTPUT · PACK KENNAMETAL_WIDIA V0.10.0

CNMG 120408-49 - TN2000

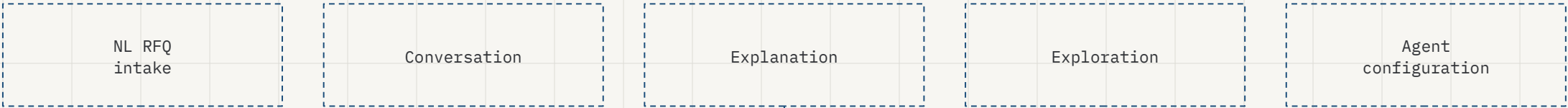


The eight character spans are the engine's own output.

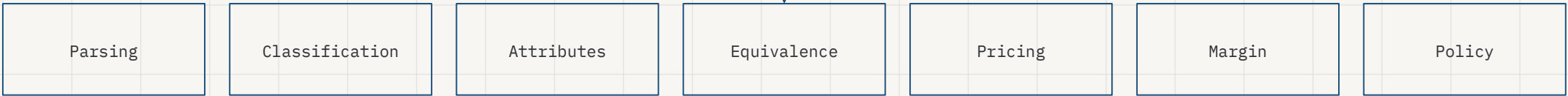
Correctness is deterministic.

Ambiguity is where AI earns its place.

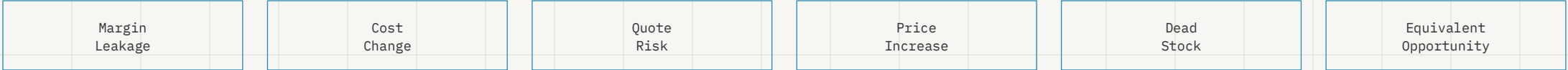
AI LAYER – OPTIONAL



DETERMINISTIC ENGINE



DETERMINISTIC WATCHERS



The AI never computes a number. Every figure it states must trace to a supplied fact or the response is rejected and the deterministic reading is shown.

The AI ships off by default — the engine underneath is the valuable half.

The engine is manufacturer-agnostic by construction

CATEGORY LADDER

Cutting tools
Industrial tools
Bearings
Electrical
MRO
Automation
Fasteners
Consumables

A new manufacturer is a new data pack. Zero manufacturer literals in the engine — tested, not asserted.

SIZED OPPORTUNITY

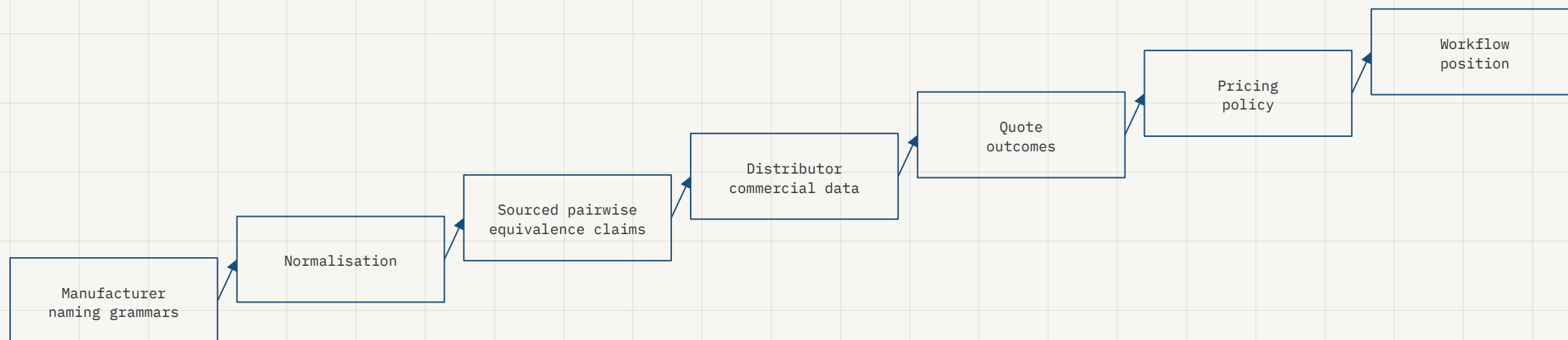
TAM	=	A × B
SAM	=	TAM × C
SOM	=	SAM × D
A	[ASSUMPTION]	distributors in scope
B	[ASSUMPTION]	annual spend each
C	[ASSUMPTION]	share on a connected ERP
D	[ASSUMPTION]	reachable in three years

The layer between describing a product and quoting it

DASHED HERE = INFORMAL, NOT AI

ERP	records the transaction
PIM	describes the product
PIE	understands it and recommends the decision
CPQ	configures and issues the quote
PRICING PLATFORMS (PRICEFX)	optimise price
AI COPILOTS	converse about it
SPREADSHEETS + MEMORY	the actual incumbent
REPLACES NOTHING	It supplies the layer none of them owns.

The asset is the decoded nomenclature, and it only accumulates



COMPOUNDS WITH EVERY PACK, EVERY QUOTE

Each manufacturer's code system is reverse-engineered once into a versioned pack: 11 families, 6,717 rows, all at 100%. That structure is published nowhere — not a data-scale claim, but a specific artefact that took specification work to build.

Land on the quote desk. Expand into the commercial decision.

PRICING MODEL

Platform subscription per organization

[ASSUMPTION]

Quote-desk seats

[ASSUMPTION]

SKU intelligence volume

[ASSUMPTION]

Premium modules

[ASSUMPTION]

The quote desk is the wedge because it is where a distributor already feels the pain daily and where the value is legible in one session. Price points are unvalidated.

LAND AND EXPAND



Current Stage

BUILT

Parser — 11 families,
6,717 rows, 100%

Platform — 2,895 tests

7 ERP connectors

Role-gated economics

IN PROGRESS

Second manufacturer pack

Outcome Tracker beyond
four detectors

VALIDATED

Live Zoho data · three entities

Coverage: 21.6% of 15,028
items; 42.5% stocked Kennametal

Determinism byte-verified

UNVALIDATED

Willingness to pay

Time saved per quote

Margin impact

External customer

NO REVENUE · NO PARTNERS · NO PIPELINE

The thesis in five answers

WHY THIS PROBLEM

The quote is where margin is made, and it runs on memory.

WHY NOW

Distributor ERPs are readable by API; models handle the ambiguous edge.

WHY PIE

Deterministic, auditable, with AI confined to where it helps.

WHY PIE WINS

The pack asset compounds; the engine is manufacturer-agnostic.

WHY THIS GETS LARGE

The same engine crosses categories without new code.